Parametric Noise Modeling Framework

Component-Physics Architecture, Equations, Output Logic and Research Gaps

A source-grounded implementation paper for PhysicsNPDModel

Implementation snapshot	Scope
28 July 2026	Conceptual aircraft screening; SEL and LAmax only

Purpose. Explain how aircraft and engine specifications become component spectra, receiver time histories, event metrics and ANP-style Noise-Power-Distance tables. The paper distinguishes implemented equations from their literature families and identifies the evidence still required.

PNMF is a research and design-screening framework. Its output is not a certified noise level, certification prediction, airport contour approval, or substitute for manufacturer engine decks and measured source data.

Abstract

PNMF contains two intentionally independent routes to ANP-style NPD tables. The learned route fits Extra Trees or Random Forest models to aircraft noise and performance data. The physics route implemented by PhysicsNPDModel instead evaluates semi-empirical component sources in 24 one-third-octave bands, propagates them to a receiver, constructs an A-weighted time history, and integrates that history into LAmax and SEL. The routes exchange no features, targets or fitted parameters; only their final outputs can be compared.

The current physics implementation resolves wing trailing-edge, slat, flap main-edge, flap side-edge, nose landing gear and main landing gear sources. Propulsion contains an explicit low-fidelity mixed-jet and fan fallback, a gated Stone-style multi-stream option, a gated Heidmann-style fan option, and an optional core/combustor source. Every physical input can be marked supplied, estimated or unavailable. Event diagnostics retain source enablement, component LAmax/SEL, receiver histories and unmodelled effects.

The implementation is structurally tested, but the newly detailed Stone- and Heidmann-style paths are not yet absolute engine-deck predictions. They share anchor constants fitted using the legacy A320-211 fallback configuration. The central research gap is therefore not additional code alone: it is traceable engine-deck data, source-level measurements, installation models and a staged external validation campaign.

Keywords: aircraft noise, conceptual design, NPD, Stone, Heidmann, Fink, SEL, LAmax, ANP, Doc 29.

1. Problem Definition and Scientific Boundary

Aircraft noise contour workflows usually consume NPD relationships indexed by engine power and slant distance. EASA makes ANP information available for noise-modelling purposes, and ECAC Doc 29 describes the wider calculation framework [R1-R2]. A future aircraft concept has no certified NPD dataset. PNMF addresses that early-design gap by generating NPD-equivalent screening tables from parametric aircraft information.

The physics route answers a narrow question:

> Given a conceptual configuration, an instantaneous flight state and explicit source assumptions, what A-weighted free-field SEL and LAmax trends follow from the implemented component scaling laws?

It does not currently answer:

- What certified EPNL or PNLTM will the aircraft achieve?
- What airport contour results after shielding, ground, terrain, lateral attenuation and non-uniform weather?
- What exact source spectrum will a new engine produce without an engine deck or measurements?
- What statistically calibrated physics uncertainty interval applies?

These boundaries matter because an additive source anchor cannot legitimately stand in for missing installation or propagation physics. PNMF therefore lists excluded effects in the event diagnostics rather than absorbing them silently.

2. Architecture: Inputs to Outputs

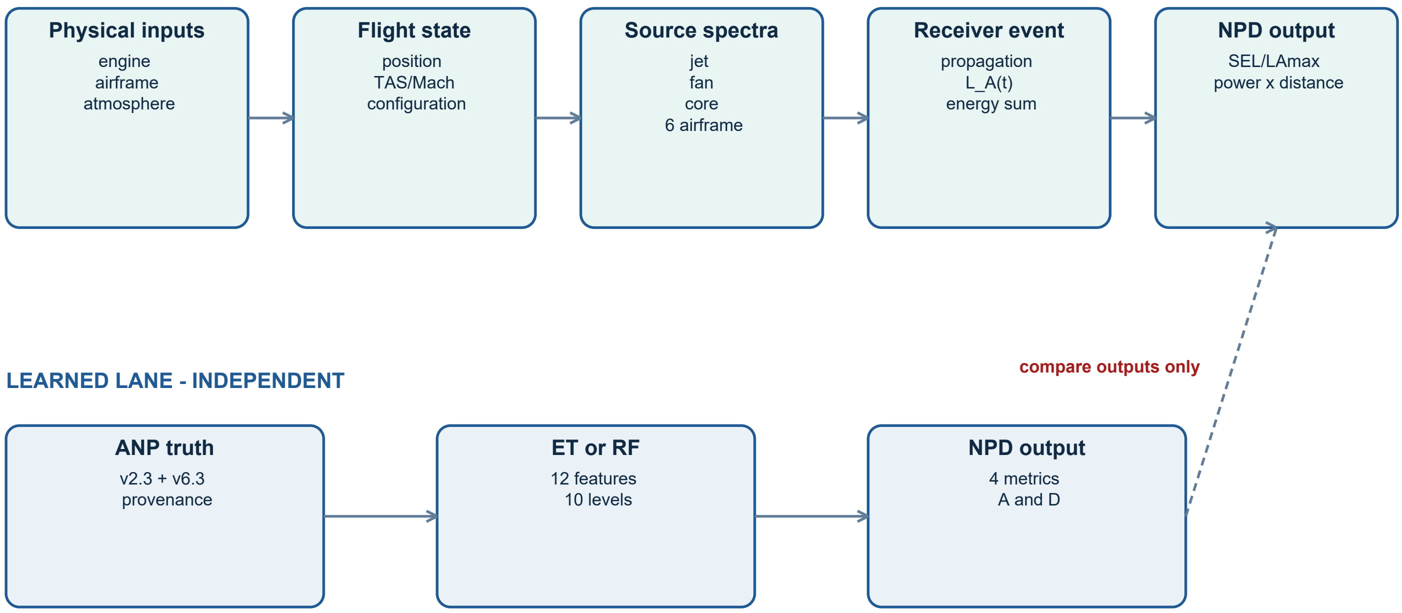

Figure 1. The physics and learned lanes produce compatible NPDTable objects but remain independent. The dashed connection is an output comparison, not a training or calibration path.

The physics data path contains seven transformations:

Stage	Primary object	Transformation	Inspectable output
1	PhysicalInput[T]	Attach supplied / estimated / unavailable evidence	Input status map
2	PhysicsDesign	Apply legacy-compatible geometry and engine adapters	Resolved design/configuration
3	FlightState	Represent instantaneous position, speed, attitude, thrust and configuration	Trajectory sample
4	Source models	Evaluate component one-third-octave spectra and directivity	24-band spectrum per component
5	Propagation	Apply spherical spreading, atmospheric absorption and A-weighting	Receiver spectrum per component
6	Event integration	Energetically sum bands/components through time	Component and total L_A(t), LAmax, SEL
7	NPD emission	Repeat over power settings and ten ANP distances	NPDTable for SEL or LAmax

The standard adapter samples 69 emission angles from 5 to 175 degrees for a level 160 kt flyover. A future trajectory module can implement the same FlightStateSource boundary without rewriting the component models.

3. Physical Input Contracts and Evidence States

A number is not automatically a high-fidelity physical input. PhysicalInput[T] therefore carries a value, one of three evidence states, and an optional note. Available estimated values may drive a model, but diagnostics must not label them as supplied.

Contract	Representative fields	Why the fields matter
EnginePhysicalInputs	thrust, BPR, mass flow, nozzle A/V/T/p, fan D, RPM/N1, blade/stator counts, spacing, fan dT, core/combustor state	Gates detailed jet, fan and core paths
AirframePhysicalInputs	wing/span, flap/slat area/chord/deflection, nose/main wheels and struts, gear state	Separates six airframe sources
AtmosphericPhysicalInputs	temperature, relative humidity, pressure	Controls frequency-dependent absorption
FlightTrajectoryInputs	position, TAS/Mach, altitude, attitude, thrust, configuration	Converts to instantaneous FlightState

Legacy compatibility. Existing PhysicsDesign calls remain valid. When geometry is absent, wing area is estimated from a 600 kg/m2 wing loading and span from aspect ratio 9. Those values are marked estimated. Missing jet streams, fan deck or complete core state are marked unavailable.

No hidden fidelity promotion. A single typed nozzle state can improve the simplified mixed-jet path, but it cannot create outer, inner and merged streams. Detailed multi-stream operation still requires explicit stream objects.

4. Acoustics Foundation

All sources return sound-pressure levels at a 1 m free-field reference in the 24 one-third-octave centre bands from 50 Hz to 10 kHz. The band set matches the ANP spectral-class range. Sound levels are combined energetically, never arithmetically.

For band levels L_b, the A-weighted total is:

```
L_A = 10 log10( sum_b 10^((L_b + A(f_b))/10) )
```

The implemented analytic A-weighting follows the standard response form and is normalised to approximately 0 dB at 1 kHz [R6]:

```
R_A(f) = (12194^2 f^4) / [ (f^2+20.6^2) sqrt((f^2+107.7^2)(f^2+737.9^2)) (f^2+12194^2) ]
A(f) = 20 log10(R_A(f)) + 2.00
```

Atmospheric absorption alpha(f,T,RH,p) is evaluated with the oxygen and nitrogen relaxation-frequency formulation in ISO 9613-1 [R5]. PNMF defaults to 15 degC, 70% relative humidity and 101.325 kPa. Typed event atmosphere inputs override those defaults.

```
alpha = 8.686 f^2 [ 1.84e-11 (p0/p) (T/T0)^0.5
+ (T/T0)^(-2.5) { 0.01275 exp(-2239.1/T)/(fr,O + f^2/fr,O)
+ 0.1068 exp(-3352.0/T)/(fr,N + f^2/fr,N) } ] [dB/m]
```

This is atmospheric absorption only. It is not a ground, terrain, shielding or lateral-attenuation model.

5. Legacy Engine-State Adapter

When a component deck is unavailable, EngineState.from_design maps thrust and BPR to a visible low-fidelity state. These equations preserve the pre-existing screening route and provide continuous trends; they are not manufacturer cycle data.

```
v_j,max = 700 / (1 + BPR)^0.44
v_j = v_j,max sqrt(F/F_max)
m_dot = F / v_j
A_j = m_dot / (rho_j v_j), rho_j = 0.64 rho_0
d_j = sqrt(4 A_j / pi)
M_tip,max = 1.6 / (1+BPR)^0.15
M_tip = M_tip,max (F/F_max)^0.45
```

If fan diameter is absent, it is inferred from maximum mass flow using a Mach-0.45 fan-face assumption. Blade-passing frequency is:

```
BPF = M_tip c_0 N_blades / (pi D_fan)
```

Quantity	Supplied path	Fallback path	Diagnostic consequence
Nozzle V/A/T/p	Typed or stream data	Thrust/BPR estimate	simplified jet: supplied or estimated
Fan flow/dT/RPM/counts/ spacing	FanDeck or complete typed set	m_dot, M_tip and default blades	Heidmann or simplified fan
Wing and high-lift geometry	AirframePhysicalInputs	weight/configuration rules	supplied or estimated airframe
Core state and attenuation	Complete CoreStream	none	source present or unavailable

6. Jet Source Logic

The simplified mixing-noise path follows a Lighthill/Stone-family V^8 sensitivity [R7]. The implemented band anchor is:

```
L_jet,1m = C_jet + 80 log10(v_j/c_0) + 10 log10(A_j) + D_jet(theta)
f_peak = f_scale (0.25 v_j/d_j)
```

The smooth band shape is a parabolic haystack in log2(f/f_peak), limited by a -40 dB floor. The tabulated directivity is aft-dominant.

Detailed multi-stream gate. The detailed option requires at least valid outer and merged streams. Each stream supplies velocity, mass flow, diameter, temperature and total pressure. Inner and intermediate streams are optional.

```
L_s = C_jet + 80 log10(v_s/c_0) + 10 log10(A_s)
+ 20 log10(288.15/T_s) + D_s(theta)
L_jet,b = 10 log10( sum_s 10^(L_s,b/10) )
```

Separate outer, inner/intermediate and merged virtual-source spectra retain different peak widths and directivity offsets. This architecture follows the multi-region idea in Stone's coaxial-jet work [R8-R9], but the equations above are PNMF's current reduced implementation - not a complete reproduction of NASA ST2JET.

If the gate fails, PNMF uses the simplified mixed-jet path and records that fallback. It does not fabricate high-fidelity streams from thrust and BPR.

7. Fan and Core Source Logic

The legacy fan fallback uses mass flow and tip Mach:

```
L_fan,1m = C_fan + 10 log10(m_dot) + 40 log10(M_tip) + D_fan(theta)
f_peak = f_scale BPF
```

A single +8 dB BPF-band tone is added in the fallback. This remains an estimated trend model.

The detailed Heidmann-style path is enabled only when mass flow, temperature rise, tip speed, fan diameter, blade and stator counts, rotor-stator spacing, and RPM or N1 evidence are present. The implementation uses:

```
L_H = C_fan + 10 log10(m_dot) + 20 log10(Delta T_t)
- 3 log10(1 + spacing/D_fan) + D_inlet/discharge(theta)
```

- Separate inlet and discharge spectra/directivities.
- BPF harmonics 1, 2 and 3 with +8, +5 and +3 dB band additions.
- Buzz-saw eligibility only when fan tip Mach is at least 1.0.
- Estimated complete inputs remain labelled estimated and are not called engine-deck-equivalent.

Heidmann's original method predicts broadband, discrete-tone and combination-tone components and depends on mass flow, temperature rise, tip relative Mach and installation details [R10]. PNMF presently implements a reduced Heidmann-style structure, not the full NASA TM-X-71763 correlation set.

Optional core/combustor. The component is absent unless core flow, exhaust geometry/state, combustor exit temperature, pressure and turbine attenuation are complete:

```
L_core = C_jet - 8 + 80 log10(v_core/c_0) + 10 log10(A_core)
+ D_jet(theta) - A_turbine
```

This core law is a provisional screened component and requires a dedicated combustor/turbine source formulation before research-grade use.

8. Six-Component Airframe Decomposition

The airframe source family follows Fink-style semi-empirical velocity and geometry scaling [R11]. The implementation reports every spectrum separately.

```
c_bar = S_wing / b
Re = rho_0 V c_bar / mu
delta_star = 0.37 c_bar Re^(-0.2)
D_dipole(theta) = 10 log10(sin^2(theta) + 0.05)
```

Component	Implemented level scaling	Peak-frequency rule
Wing trailing edge	C_wf + 50 log10(V/c0) + 10 log10(delta_star b) + D_dipole	0.1 V / delta_star
Slat	Separate deployed spectrum based on wing level and slat chord	0.2 V / c_slat
Flap main edge	C_wf + 3 + 50 log10(V/c0) + 10 log10(S_f sin^2(delta_f)); 75% energy	0.6 V / (0.3 c_bar)
Flap side edge	Same parent law; 25% energy with shifted spectral shape	1.8 x main-edge peak
Main landing gear	C_g + 60 log10(V/c0) + 10 log10(n_m d_m^2) + strut factor	0.8 V / d_m
Nose landing gear	C_g + 60 log10(V/c0) + 10 log10(n_n d_n^2) + strut factor	1.25 x 0.8 V / d_n

```
strut factor = 10 log10(1 + d_strut^2 / d_wheel^2)
L_airframe,b = 10 log10( sum_k 10^(L_k,b/10) )
```

The flap 75/25 energy split and the selected spectral multipliers are explicit PNMF implementation choices that preserve the legacy aggregate energy while making components inspectable. They require validation against isolated component spectra before being interpreted as universal Fink correlations.

9. Propagation, Receiver Time History and Event Metrics

For a component band level at the 1 m reference, PNMF applies:

```
L_b(r) = L_b(1 m) - 20 log10(r) - alpha_b r + A(f_b)
L_engine,b,total = L_engine,b,one + 10 log10(N_engines)
```

At each receiver time sample, band and component energies are summed:

```
L_k,A(t) = 10 log10( sum_b 10^(L_k,b(t)/10) )
L_A(t) = 10 log10( sum_k 10^(L_k,A(t)/10) )
```

For the reference flyover, closest slant distance is d, emission angle is theta, and speed V is 160 kt:

```
r(theta) = d / sin(theta)
x(theta) = -d / tan(theta)
t(theta) = x(theta) / V
```

The event metrics are then:

```
LAmax = max_t L_A(t)
SEL = 10 log10( integral 10^(L_A(t)/10) dt / 1 s )
```

The distance-dependent SEL duration effect emerges from the receiver history; it is not appended as a separate correction. EventDiagnostics retains the total and component histories, component LAmax/SEL, enablement/fallback status, input evidence and excluded effects.

Only free-field spreading and frequency-dependent absorption are included. Shielding, nacelle treatment, ground reflection, lateral attenuation, terrain and non-uniform atmosphere are explicit future modules.

10. NPD Output Logic and Calibration Policy

PhysicsNPDModel repeats the reference event over requested per-engine thrust settings and the ten standard ANP distances: 200, 400, 630, 1,000, 2,000, 4,000, 6,300, 10,000, 16,000 and 25,000 ft. The result is an NPDTable with the same structural interface used by the learned route.

Property	Physics route
Metrics	SEL and LAmax only
Operating modes	Approach and departure configuration adapters
Internal units	metres, newtons, m/s, kelvin/degC, pascals/kPa
NPD boundary units	feet and lbf per engine
Distance interpolation	linear level interpolation in log10(distance)
Power interpolation	linear level interpolation in power
Truth/prediction storage	strictly separate tables

Frozen calibration. Four additive source anchors and one shared spectral placement factor are fitted by bounded least squares to one reference aircraft, A320-211 at BPR 6.0, using SEL and LAmax for both modes:

```
c* = arg min_c sum_(metric,mode,power,distance)
[ L_physics(c) - L_ANP,A320-211 ]^2
c = [C_jet, C_fan, C_wingflap, C_gear, log2(f_scale)]
```

After calibration, the coefficients are frozen for all other aircraft. This policy prevents per-aircraft tuning and keeps fleet checks out of sample.

Important limitation: calibration currently exercises the legacy fallback configuration because the A320 reference path does not supply complete detailed stream/fan/core decks. The detailed Stone- and Heidmann-style paths reuse these anchors but have not been separately calibrated or externally validated.

11. Diagnostics, Tests and Current Validation Evidence

Figure 2. Validation must climb from evidence and source behavior to external measurements. Passing a higher software-level test does not replace a lower physical-data requirement.

Layer	Current evidence	What it establishes	What remains
Inputs	Unit tests for evidence states, typed input consumption and core completeness	Contracts are active and fallbacks visible	Independent data-quality audit
Sources	Scaling, directivity, six-component decomposition, detailed gates	Implemented behavior and energetic arithmetic	Primary-correlation reproduction and rig spectra
Receiver history	69-sample timing, energetic total, duration trend	Event integration is internally consistent	Measured flyover histories
NPD tables	SEL/LAmax scope, shape, monotone distance behavior, backward compatibility	Downstream interface remains usable	Detailed-model fleet validation
Repository	20 focused physics tests; full documented suite 78 passed, 1 skipped	Software regression protection	Certification-quality verification is out of scope

Historical project artifacts report a frozen-physics fleet median RMSE of 2.82 dB across 12 aircraft and an A320 calibration RMSE of 1.57 dB. Those figures belong to the earlier aggregate/fallback implementation and were not rerun as part of the component-resolution change. They must not be used as validation of the new detailed jet/fan/core branches.

Tree dispersion from ET/RF is a learned-ensemble disagreement heuristic. It is not component model-form uncertainty and is not a calibrated physics interval.

12. Research Gaps and Required Work

The remaining gaps are prioritised by scientific dependency, not coding convenience.

Priority	Gap	Required evidence / implementation	Acceptance criterion
P0	Engine-deck provenance	Per-stream nozzle area, V, T, p, mass flow; fan maps, RPM/N1, blade/stator geometry; core state	No detailed source marked supplied without traceable deck origin
P0	Source-law reproduction	Reproduce selected Stone, Heidmann and Fink reference cases before PNMF-specific simplification	Band spectra/directivity agree within predeclared tolerances
P0	Detailed-source calibration	Separate anchor policy or absolute constants using non-A320 development data	Frozen validation set remains untouched
P1	Component measurements	Static rig and flight spectra for jet, fan and airframe configurations	Component and total residuals reported by band/angle/power
P1	Installation physics	Shielding, nacelle liner/suppression, pylon/wing interference	No source anchor absorbs installation deltas
P1	Propagation	Ground reflection, lateral attenuation, terrain and non-uniform atmosphere	Verification against Doc 29 cases and measured histories
P1	Trajectory integration	General segment path, Doppler/convective effects, airport geometry	Receiver histories verified before contours
P2	Perceived-noise metrics	PNL, tone corrections, PNLTM and EPNL duration integration	Independent benchmark suite; no derivation from A-weighted metrics
P2	Uncertainty	Input distributions, model discrepancy, calibration/posterior validation	Coverage checked on held-out physical measurements
P2	Novel configurations	Distributed propulsion, BWB shielding, open rotor/electric source families	Configuration-specific source and installation evidence

Recommended next experiment. Select one conventional turbofan with a public or partner-supplied deck and source spectra. Freeze all equations and tolerances before evaluating it. Validate outer/inner/merged jet and inlet/discharge fan spectra at several operating points, then validate the receiver time history, and only then compare SEL/LAmax NPD tables.

13. Recommended Development Sequence

Phase	Work package	Deliverable	Stop condition
A	Reference-case harness	Machine-readable source cases from cited reports	No source upgrade until baseline equations reproduce cases
B	One-engine deck integration	Traceable EnginePhysicalInputs with field-level evidence	Missing deck fields remain unavailable, never synthesized silently
C	Source validation	Spectrum/directivity residual plots by component	Investigate bias before event fitting
D	Receiver validation	Measured-vs-model L_A(t) and band histories	No SEL/LAmax tuning before time-history agreement
E	Frozen external set	Aircraft/engine/configuration split fixed in advance	No calibration leakage
F	Installation/ground modules	Independent corrections and verification cases	Source constants remain unchanged
G	Tone/perceived-noise route	PNLT/PNLTM/EPNL implementation and tests	Physics API remains SEL/LAmax-only until complete

This sequence preserves the most important architecture invariant: a correction belongs to the physical mechanism that causes it. Source calibration, installation, propagation and metric conversion remain separate modules.

14. Conclusions

PNMF now has a trajectory-ready, component-resolved physics architecture with explicit evidence states and inspectable output logic. Aircraft and engine inputs become source spectra; source spectra become propagated component histories; histories become LAmax and SEL; repeated events become ANP-style NPD tables. Six airframe components, detailed jet virtual sources, fan inlet/discharge branches and optional core noise can be examined independently.

The framework is scientifically useful because it exposes causal sensitivities and failure modes that a learned model cannot. Its present limitation is equally clear: the detailed branches are structurally implemented but not yet supported by a complete engine-deck and component-measurement validation chain. The correct next step is disciplined validation and data acquisition, not stronger accuracy language.

Until the gaps in Section 12 are closed, PhysicsNPDModel should be described as an independent conceptual-screening cross-check for SEL and LAmax.

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Reference interpretation. PNMF uses Stone-, Heidmann- and Fink-style families as scientific foundations. Unless explicitly stated, the equations in this paper document the current PNMF implementation and should not be cited as exact reproductions of the original reports.